

Statistics II: Introduction to Inference

Problem set 7

1. Show that the following distributions belong to either location, or scale, or location scale families. Hence find the appropriate transformation of a random variable belonging to these distributions, such that the transformed random variable has a distribution free of the parameters θ .
 - (a) `normal`(μ, σ^2),
 - (b) `exponential`(λ),
 - (c) `uniform`($-\theta, \theta$),
 - (d) `exponential`(λ),
 - (e) `Gamma`(n, θ), with n known,
 - (f) `normal`(θ, θ^2),
 - (g) `uniform`($\theta - 1/2, \theta + 1/2$).
2.
 - (a) Let $X_1, \dots, X_n$ be a random sample of size n from a location family of distribution with location parameter θ . Show that the distribution of any function of $Y_{i,j} = \{X_i - X_j\}$, $i, j = 1, \dots, n$ is free of θ .
 - (b) Let $X_1, \dots, X_n$ be a random sample of size n from a scale family of distribution with scale parameter θ . Show that the distribution of any function of $Y = X_1^2 / \sum_{j=1}^n X_j^2$ and $Z = X_{(1)} / X_{(n)}$ are free of θ .
3. The independent random variables $X_1, \dots, X_n$ have common distribution

$$P(X_i \leq x) = \begin{cases} 0 & \text{if } x \leq 0 \\ (x/\beta)^\gamma & \text{if } 0 < x < \beta \\ 1 & \text{if } x \geq \beta \end{cases}.$$

Find a $(1 - \alpha)100\%$ confidence interval for β based on the MLE of β , when γ is known.

4. Find a $(1 - \alpha)$ confidence interval for θ , given $X_1, \dots, X_n$ iid with pdf
 - (a) $f_X(x; \theta) = 1, \theta - 0.5 \leq x \leq \theta + 0.5$.
[Instead of the exact confidence interval, you may find a approximate confidence interval based on the asymptotic distribution of an appropriate pivot.]
 - (b) $f_X(x; \theta) = 2x\theta^{-2}, 0 < x < \theta, \theta > 0$.
5. Let $X_1, \dots, X_n$ be iid from `Uniform`($\theta, 1$) distribution, $\theta < 1$.
 - (a) Obtain a suitable pivot for finding a confidence interval for θ .
 - (b) Based on this test find a $(1 - \alpha)$ confidence set.
6. Let T be a statistic with continuous strictly decreasing CDF $F_T(\cdot; \theta)$, and α_1, α_2 be such that $\alpha_1 + \alpha_2 = \alpha$, for some fixed $\alpha \in (0, 1)$. Suppose that for each t the functions $L(t)$ and $U(t)$ are defined as

$$F_T(T; U(T)) = \alpha_1, \quad \text{and} \quad F_T(T; L(T)) = 1 - \alpha_2.$$

Then show that the random interval $[L(t), U(t)]$ is a $(1 - \alpha)$ confidence interval for θ .

7. Let X be a single observation from a `beta`($\theta, 1$) distribution. Let $Y = -(\log X)^{-1}$. Evaluate the confidence coefficient, β , of the interval $[y/2, y]$.
8. A confidence interval $[L(\mathbf{X}), U(\mathbf{X})]$ for the parameter θ with confidence coefficient at least $1 - \alpha$ is called *unbiased* if $P_\theta(L(\mathbf{X}) < \theta < U(\mathbf{X})) \geq 1 - \alpha$, and $P_\theta(L(\mathbf{X}) < \theta' < U(\mathbf{X})) \leq 1 - \alpha$ for all $\theta' \neq \theta$.
Based on a random sample of size n from `uniform`($0, \theta$), show that the symmetric confidence interval obtained from the pivot $X_{(n)}/\theta$ is unbiased for sufficiently large n .
9. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu_0, \sigma^2)$ where μ_0 is known.
 - (a) Find the UMVUE of σ^2 .
 - (b) Using the UMVUE, find an appropriate pivot for σ^2 , and its distribution.
 - (c) Using this pivot find a $(1 - \alpha)$ -confidence interval for σ^2 .
10. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{exponential}(\theta)$.
 - (a) Find an appropriate transformation $W_i = T(X_i, \theta)$ such that the distribution of W_i does not depend on $1/\theta$.
 - (b) Consider two pivots, one based on W_1 only, and another involving the UMVUE of $1/\theta$.
 - (c) Find the symmetric $(1 - \alpha)$ -confidence intervals using the two pivots obtained in part (b).
 - (d) Generate $n = 10$ IID samples from `exponential`(5) distribution. Based on the samples obtain realizations of the two confidence intervals for $\alpha = 0.1, 0.05, 0.025$.